Statistical Estimation Theory

Point Estimation, Confidence Intervals, and Applications

Comprehensive Lecture Notes

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Contents

1 Introduction to Statistical Estimation

1.1 The Core Problem in Statistics

In statistics and data science, we often face a fundamental challenge: **We want to know something about a large group (population), but we can only examine a small part of it (sample).**

DefinitionPopulation and Sample

A **population** is the complete set of all items or individuals of interest in a study. A **sample** is a subset of the population selected for observation and analysis.

1.1.1 Real-World Examples

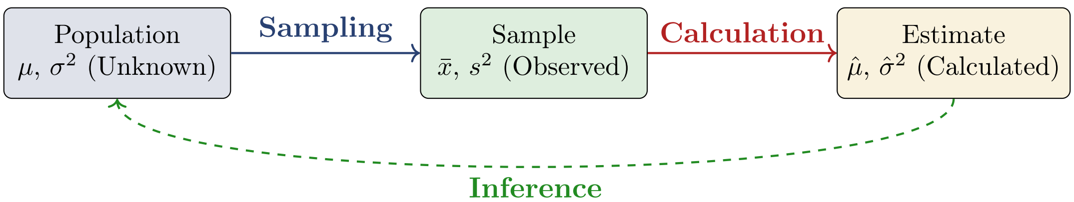
- **Market Research:** Want to know average income of all customers, but can only survey 1,000
- **Quality Control:** Need to know defect rate of all products, but can only test 100
- **Medical Studies:** Want to understand effect of drug on all patients, but can only study 500
- **Political Polling:** Need to predict election results but can only poll a few thousand voters

1.2 Population Parameters vs Sample Statistics

Theorem/ConceptParameters vs Statistics

Population Parameter	**Sample Statistic**
Numerical characteristic of entire population	Numerical characteristic of sample
Denoted by Greek letters (μ, σ, π)	Denoted by Latin letters ($\bar{x}$, s, p)
Usually unknown and fixed	Calculated from sample data
Example: Population mean μ	Example: Sample mean $\bar{x}$

1.3 The Estimation Process



2 Fundamental Concepts

2.1 What is Estimation?

DefinitionStatistical Estimation

Estimation is the process of using sample data to:

1. Infer or approximate unknown population parameters
2. Make educated judgments about population characteristics
3. Quantify uncertainty in our inferences

Key IdeaCharacteristics of Estimation

- **Not Exact:** Estimates are approximations, not precise values
- **Based on Probability:** Incorporates concepts of randomness and chance
- **Quantifies Uncertainty:** Provides measures of reliability (confidence levels)
- **Fundamental to Inference:** Basis for statistical inference and decision making

2.2 Estimator vs Estimate

Theorem/ConceptKey Distinctions

Term	Definition	Example
Estimator	Rule/formula for calculating estimate	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Estimate	Numerical value obtained from estimator	$\bar{x} = 175$ cm
Parameter	True population value (unknown)	Population mean μ

2.3 Properties of Good Estimators

DefinitionDesirable Properties

An estimator $\hat{\theta}$ is said to be:

- **Unbiased:** if $\mathbb{E}(\hat{\theta}) = \theta$ (correct on average)
- **Consistent:** if $\hat{\theta} \xrightarrow{p} \theta$ as $n \to \infty$ (improves with more data)
- **Efficient:** if it has minimum variance among all unbiased estimators (most precise)
- **Sufficient:** if it uses all information in the sample about θ (no information waste)

3 Point Estimation

DefinitionPoint Estimate

A **point estimate** is a single numerical value used as the "best guess" or "best estimate" of an unknown population parameter.

3.1 Common Point Estimators

Population Parameter	**Point Estimator**	**Formula**
Mean (μ)	Sample Mean ($\bar{x}$)	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Variance (σ^2)	Sample Variance (s^2)	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
Proportion (p)	Sample Proportion ($\hat{p}$)	$\hat{p} = \frac{x}{n}$ where x = successes
Standard Deviation (σ)	Sample SD (s)	$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

ExampleEstimating Average Height

Problem: Estimate the average height of university students.
Data: A sample of 50 students has mean height = 168.5 cm
Solution:

$$\begin{aligned}\text{Population parameter: } & \mu \text{ (true average height)} \\ \text{Point estimator: } & \bar{x} = 168.5 \text{ cm} \\ \text{Interpretation: } & \text{Our best guess for } \mu \text{ is } 168.5 \text{ cm}\end{aligned}$$

Limitation: We don't know how close 168.5 cm is to the true average. This is the fundamental weakness of point estimation.

3.2 Limitations of Point Estimation

- **No Precision Information:** Doesn't indicate how close the estimate might be to the true value
- **Sample Variability:** Different samples give different estimates
- **Overly Precise:** Presents estimate as exact value when it's approximate
- **Poor Decision Support:** Difficult to base decisions on single value without confidence measure

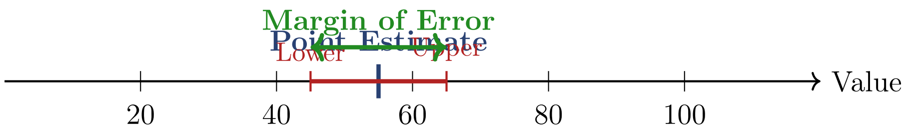
4 Interval Estimation

DefinitionInterval Estimate

An **interval estimate** is a range (or interval) of values that is likely to contain the unknown population parameter.

4.1 General Form of Interval Estimate

$$\text{Interval Estimate} = \text{Point Estimate} \pm \text{Margin of Error}$$



4.2 Advantages Over Point Estimation

Key IdeaWhy Use Intervals?

- **Quantifies Uncertainty:** Shows range of plausible values
- **Accounts for Sampling Error:** Recognizes sample variability
- **Better Decision Support:** Provides basis for risk assessment
- **Comparative Analysis:** Allows overlap/non-overlap assessment
- **More Honest Communication:** Represents what we actually know

5 Confidence Intervals

DefinitionConfidence Interval

A **confidence interval** is an interval estimate constructed so that, with a specified probability (confidence level), it contains the true population parameter.

Theorem/ConceptInterpretation of 95% Confidence Interval

"If we repeated our sampling process many times and constructed a 95% CI from each sample, about 95% of those intervals would contain the true population parameter."

Critical Warning: For any specific calculated interval, the true parameter is either in it or not. The 95% refers to the **method**, not the probability that our specific interval contains the parameter.

5.1 General Formula for Confidence Interval of Mean

$$\text{CI} = \bar{x} \pm Z_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

where:

- $\bar{x}$ = sample mean (point estimate)
- $Z_{\alpha/2}$ = critical value from standard normal distribution
- s = sample standard deviation
- n = sample size
- $\frac{s}{\sqrt{n}}$ = standard error (SE)

5.2 Critical Z-Values for Common Confidence Levels

Confidence Level	α	**Z-Critical Value** ($Z_{\alpha/2}$)
80%	0.20	1.282
85%	0.15	1.440
90%	0.10	1.645
95%	0.05	1.960
98%	0.02	2.326
99%	0.01	2.576

ExampleDetailed CI Calculation

Problem: Construct a 95% confidence interval for the average height of city residents.
Given:

- Sample size: $n = 40$
- Sample mean: $\bar{x} = 175$ cm
- Sample standard deviation: $s = 20$ cm
- Confidence level: 95% $\Rightarrow Z = 1.96$

Step 1: Calculate Standard Error (SE)

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{20}{\sqrt{40}} = \frac{20}{6.3246} \approx 3.162 \text{ cm}$$

Step 2: Calculate Margin of Error (MOE)

$$\text{MOE} = Z \times \text{SE} = 1.96 \times 3.162 \approx 6.20 \text{ cm}$$

Step 3: Construct Confidence Interval

$$\text{Lower bound} = \bar{x} - \text{MOE} = 175 - 6.20 = 168.80 \text{ cm}$$
$$\text{Upper bound} = \bar{x} + \text{MOE} = 175 + 6.20 = 181.20 \text{ cm}$$

Final Result: $95\%\text{CI} = (168.80 \text{ cm}, 181.20 \text{ cm})$
Interpretation: We are 95% confident that the true average height of all city residents lies between 168.80 cm and 181.20 cm.

5.3 Factors Affecting Confidence Interval Width

Factor	Change	Effect on CI Width
Sample Size (n)	Increase	**Decreases** (more precise)
Standard Deviation (s)	Increase	**Increases** (more variability)
Confidence Level	Increase	**Increases** (more conservative)

6 Margin of Error

DefinitionMargin of Error

The **margin of error (MOE)** is the maximum expected difference between the true population parameter and a sample estimate of that parameter. It represents the "fudge factor" that accounts for the uncertainty in using a sample statistic to estimate the population parameter.

$$\text{MOE} = Z_{\alpha/2} \times \text{SE} = Z_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

ExampleCalculating MOE

Problem: Calculate the margin of error given:

- Confidence level: 99%
- Standard deviation: $s = 23.44$
- Sample size: $n = 32$

Solution:

$$\begin{aligned}
Z \text{ for } 99\% &= 2.576 \\
\text{MOE} &= 2.576 \times \frac{23.44}{\sqrt{32}} \\
&= 2.576 \times \frac{23.44}{5.657} \\
&= 2.576 \times 4.144 \\
&\approx \mathbf{10.67}
\end{aligned}$$

Interpretation: With 99% confidence, our estimate is within ± 10.67 units of the true population value.

6.1 Sample Size Determination

Theorem/ConceptDetermining Required Sample Size

To achieve a desired margin of error:

$$\begin{aligned} \text{MOE} &= Z_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ \sqrt{n} &= Z_{\alpha/2} \times \frac{s}{\text{MOE}} \\ n &= \left(\frac{Z_{\alpha/2} \times s}{\text{MOE}}\right)^2 \end{aligned}$$

Always round up to the next whole number.

6.2 Relationship Between Sample Size and Margin of Error

Key IdeaThe Square Root Law

- **Inverse Square Root Relationship:** $\text{MOE} \propto \frac{1}{\sqrt{n}}$
- **Diminishing Returns:** To halve the MOE, you must **quadruple** the sample size:

$$n_{\text{new}} = 4 \times n_{\text{old}} \Rightarrow \text{MOE}_{\text{new}} = \frac{1}{2}\text{MOE}_{\text{old}}$$

- **Practical Implication:** Large increases in sample size yield relatively small decreases in margin of error

7 Worked Problems: Step-by-Step Solutions

7.1 Problem 1: Basic Concepts and CI Calculation

Example

Problem: A researcher wants to estimate the average time users spend on a mobile app. She samples $n = 100$ users and finds a sample mean of $\bar{x} = 8.5$ minutes.

(a) Is this a point or interval estimate?

(b) What additional information would make this estimate more useful?

(c) Assume the population standard deviation is known to be $\sigma = 3.2$ minutes. Construct a 95% confidence interval.

(d) How does the CI change if σ is unknown (use $s = 3.2$)?

(e) What if the sample size was only $n = 15$?

Solution:

Part (a): Type of Estimate

The value 8.5 minutes is a **point estimate**—a single number used as our best guess for the population mean μ.

Part (b): Needed Information

To make the estimate more useful, we need:

- **Standard deviation** (σ or s): Measures variability in user times
- **Sample size** (n): Already given as 100
- **Confidence interval**: Provides a plausible range for the true mean

Part (c): 95% CI with σ Known

Given: $\bar{x} = 8.5$, $\sigma = 3.2$, $n = 100$, $Z_{0.975} = 1.96$

$$\begin{aligned} \text{SE} &= \frac{\sigma}{\sqrt{n}} = \frac{3.2}{10} = 0.32 \\ \text{MOE} &= 1.96 \times 0.32 \approx 0.627 \\ \text{CI} &= 8.5 \pm 0.627 = (\mathbf{7.87}, \mathbf{9.13}) \text{ minutes} \end{aligned}$$

Part (d): 95% CI with σ Unknown (t-distribution)

When σ is unknown, we use the t-distribution with $df = n - 1 = 99$.
$t_{0.975,99} \approx 1.984$ (slightly larger than 1.96, reflecting extra uncertainty)

$$\begin{aligned} \text{SE} &= \frac{s}{\sqrt{n}} = \frac{3.2}{10} = 0.32 \\ \text{MOE} &= 1.984 \times 0.32 \approx 0.635 \\ \text{CI} &= 8.5 \pm 0.635 = (\mathbf{7.87}, \mathbf{9.14}) \text{ minutes} \end{aligned}$$

For large n, the difference between z and t is minimal.

Part (e): Small Sample Size ($n = 15$)

With $n = 15$, we use t-distribution with $df = 14$.
$t_{0.975,14} \approx 2.145$ (much larger due to small sample)

$$\begin{aligned} \text{SE} &= \frac{3.2}{\sqrt{15}} \approx 0.826 \\ \text{MOE} &= 2.145 \times 0.826 \approx 1.772 \\ \text{CI} &= 8.5 \pm 1.772 = (\mathbf{6.73}, \mathbf{10.27}) \text{ minutes} \end{aligned}$$

Scenario	Critical Value	MOE	95% CI
Large n, σ known (z)	1.960	0.627	(7.87, 9.13)
Large n, σ unknown (t)	1.984	0.635	(7.87, 9.14)
Small $n = 15$, σ unknown (t)	2.145	1.772	(6.73, 10.27)

ImportantKey Insight

Small sample sizes dramatically increase the margin of error and widen the confidence interval due to greater uncertainty.

7.2 Problem 2: Margin of Error and Sample Size

Example

Problem: Given $n = 64$, $\bar{x} = 120$, $s = 15$, and 90% confidence level:

(a) Calculate the 90% confidence interval

(b) Calculate the margin of error

(c) What happens to the CI if n increases to 256?

Solution:

Parts (a) and (b): CI and MOE

For 90% confidence: $Z_{0.95} = 1.645$

$$\begin{aligned} \text{SE} &= \frac{15}{\sqrt{64}} = \frac{15}{8} = 1.875 \\ \text{MOE} &= 1.645 \times 1.875 \approx \mathbf{3.08} \\ \text{CI} &= 120 \pm 3.08 = (\mathbf{116.92}, \mathbf{123.08}) \end{aligned}$$

Part (c): Effect of Increased Sample Size

With $n = 256$:

$$\begin{aligned} \text{SE}_{new} &= \frac{15}{\sqrt{256}} = \frac{15}{16} = 0.9375 \\ \text{MOE}_{new} &= 1.645 \times 0.9375 \approx \mathbf{1.54} \\ \text{CI}_{new} &= 120 \pm 1.54 = (\mathbf{118.46}, \mathbf{121.54}) \end{aligned}$$

The new interval is much narrower, reflecting greater precision. Quadrupling the sample size halved the margin of error.

7.3 Problem 3: Interpreting Survey Results

Example

Problem: A survey reports: "45% of voters support Candidate A with a margin of error of ±4% at 95% confidence."

(a) What is the confidence interval?

(b) If true support is 47%, is this consistent with the survey?

(c) What sample size was likely used? (Assume worst-case $p = 0.5$)

Solution:

Part (a): Confidence Interval

$$\text{CI} = 0.45 \pm 0.04 = (\mathbf{0.41}, \mathbf{0.49}) \text{ or } (\mathbf{41}\%, \mathbf{49}\%)$$

Part (b): Consistency Check

Since 47% (0.47) falls within the interval (41%, 49%), **yes, it is consistent** with the survey results.

Part (c): Sample Size Estimation

Using the proportion formula with worst-case $p = 0.5$ (maximizes variance):

$$\begin{aligned}
\text{MOE} &= Z \times \sqrt{\frac{p(1-p)}{n}} \\
0.04 &= 1.96 \times \sqrt{\frac{0.25}{n}} \\
\frac{0.04}{1.96} &= \sqrt{\frac{0.25}{n}} \\
0.02041 &\approx \sqrt{\frac{0.25}{n}} \\
(0.02041)^2 &\approx \frac{0.25}{n} \\
n &\approx \frac{0.25}{0.0004165} \approx 600.24
\end{aligned}$$

The likely sample size was about **601 voters** (round up).

7.4 Problem 4: Comparing Two Groups (A/B Testing)

Example

Problem: An e-commerce company tests a new checkout process:

- **Test Group:** $n_T = 200$, conversion rate $\hat{p}_T = 12.5\%$
- **Control Group:** $n_C = 200$, conversion rate $\hat{p}_C = 11.0\%$

(a) Construct 95% CIs for each group

(b) Do the intervals overlap? What does this suggest?

(c) Calculate the 95% CI for the difference in conversion rates

Solution:

Part (a): Individual Confidence Intervals

For 95% confidence: $Z = 1.96$
Test Group:

$$\mathrm{SE}_T = \sqrt{\frac{0.125 \times 0.875}{200}} \approx 0.0234$$

$$\mathrm{MOE}_T = 1.96 \times 0.0234 \approx 0.0459$$

$$\mathrm{CI}_T = 0.125 \pm 0.0459 = (\mathbf{7.9}\%, \mathbf{17.1}\%)$$

Control Group:

$$\mathrm{SE}_C = \sqrt{\frac{0.110 \times 0.890}{200}} \approx 0.0221$$

$$\mathrm{MOE}_C = 1.96 \times 0.0221 \approx 0.0433$$

$$\mathrm{CI}_C = 0.110 \pm 0.0433 = (\mathbf{6.7}\%, \mathbf{15.3}\%)$$

Part (b): Overlap Assessment

Yes, the intervals overlap (both include values around 8-15%). This suggests we cannot be confident that there is a true difference between the groups based on this data alone.

Part (c): Confidence Interval for the Difference

Point estimate of difference: $\hat{d} = 0.125 - 0.110 = 0.015$
Standard error of difference:

$$\mathrm{SE}_d = \sqrt{\frac{0.125 \times 0.875}{200} + \frac{0.110 \times 0.890}{200}} \approx 0.0322$$

Margin of error: $\mathrm{MOE}_d = 1.96 \times 0.0322 \approx 0.0631$

$$\mathrm{CI}_d = 0.015 \pm 0.0631 = (\mathbf{-0.048}, \mathbf{0.078}) \text{ or } (\mathbf{-4.8}\%, \mathbf{7.8}\%)$$

ImportantCritical Interpretation

The 95% CI for the difference contains zero. This means a difference of zero is plausible, and we **cannot conclude there is a statistically significant difference** between the test and control groups.

7.5 Problem 5: Sample Size Determination

Example

Problem: A data scientist wants to estimate average session duration ($s \approx 180$ seconds from pilot study).

(a) What sample size is needed for MOE $= \pm 30$ seconds at 95% confidence?

(b) If budget allows only $n = 100$, what MOE can be achieved?

(c) With $n = 100$, what confidence level gives MOE $= \pm 45$ seconds?

Solution:

Part (a): Required Sample Size

$$n = \left(\frac{Z \cdot s}{\text{MOE}}\right)^2 = \left(\frac{1.96 \times 180}{30}\right)^2 = (11.76)^2 \approx 138.3$$

Required sample size: **139 observations** (round up).

Part (b): Achievable MOE with $n = 100$

$$\text{MOE} = 1.96 \times \frac{180}{\sqrt{100}} = 1.96 \times 18 = \mathbf{35.28} \text{ seconds}$$

Part (c): Confidence Level for Given MOE

Find Z such that:

$$Z = \frac{\text{MOE} \cdot \sqrt{n}}{s} = \frac{45 \times 10}{180} = 2.5$$

From standard normal tables, the area between $Z = \pm 2.5$ is approximately 0.9876.
Required confidence level: **98.76%** (approximately 99%).

8 Practical Applications in Data Science & Machine Learning

ApplicationModel Performance Evaluation

- Report accuracy with confidence intervals: "Model accuracy: 92% $\pm$ 2% (95% CI: 90%-94%)"
- Use bootstrap sampling or repeated cross-validation to estimate variability
- Compare models by checking if their confidence intervals overlap

ApplicationA/B Testing and Experimentation

- Compare treatment vs control groups using confidence intervals
- Decision rule: If 95% CIs don't overlap, there's likely a significant difference
- Calculate uplift with confidence intervals: "Conversion rate increased by 2.5%$\pm$0.8%"

ApplicationSurvey Design and Analysis

- Always report polling results with margin of error
- Determine sample size based on desired precision before conducting study
- Account for design effects in complex sampling designs

Metric	Point Estimate	Confidence Interval (95%)
Customer Lifetime Value	$1,250	($1,100 - $1,400)
Churn Rate	5.2%	(4.8% - 5.6%)
Average Order Value	$45.30	($42.50 - $48.10)
Customer Satisfaction	4.2/5	(4.0 - 4.4)

9 Summary and Key Takeaways

9.1 Comparison of Estimation Methods

Aspect	Point Estimate	Interval Estimate
Definition	Single value	Range of values
Example	$\bar{x} = 175$	$(168.8, 181.2)$
Uncertainty	Not quantified	Explicitly quantified
Use Case	Quick reference	Decision making
Communication	Simple	More informative
When to Use	Preliminary analysis	Final reports, decisions

9.2 Key Formulas Summary

$$\text{Confidence Interval: } \text{CI} = \bar{x} \pm Z_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

$$\text{Margin of Error: } \text{MOE} = Z_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

$$\text{Sample Size: } n = \left(\frac{Z_{\alpha/2} \times s}{\text{MOE}}\right)^2$$

$$\text{CI} = \bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2,n-1}$ is the critical value from t-distribution with $n-1$ degrees of freedom.

$$\text{CI} = \hat{p} \pm Z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$n = \left(\frac{Z_{\alpha/2}}{\text{MOE}}\right)^2 \times p(1-p) \quad \text{(worst case: } p = 0.5\text{)}$$

9.3 Common Critical Values Reference

- 80% CL: $Z = 1.282$
- 85% CL: $Z = 1.440$
- 90% CL: $Z = 1.645$
- 95% CL: $Z = 1.960$
- 98% CL: $Z = 2.326$
- 99% CL: $Z = 2.576$

9.4 Practical Guidelines

Key IdeaBest Practices

- Always report estimates with measures of precision (MOE or CI)
- Choose confidence level based on consequence of error
- Consider trade-off between precision and cost
- Use interval estimates for important decisions
- Remember: 95% CI means the method works 95% of the time, not that there's a 95% probability the parameter is in your specific interval
- Visualize confidence intervals in reports and presentations

9.5 Common Pitfalls to Avoid

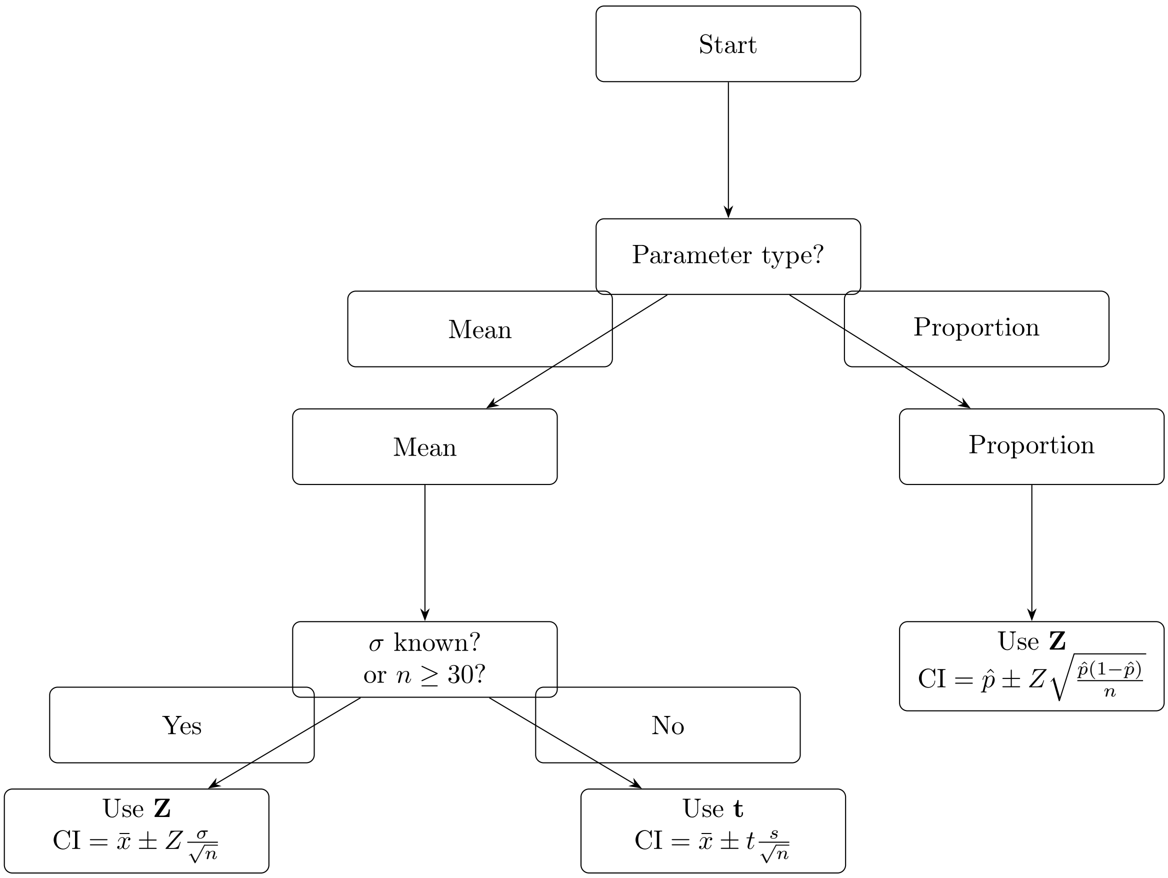
- **Misinterpretation:** A 95% CI does NOT mean "95% probability that parameter is in the interval"
- **Ignoring Assumptions:** Check normality, independence, random sampling
- **Overprecision:** Don't report too many decimal places
- **Confusing CI with Prediction Interval:** CIs are for parameters, prediction intervals are for individual observations
- **Multiple Comparison Problem:** When comparing many CIs, adjust confidence levels (Bonferroni correction)

Appendix: Quick Reference

Statistical Notation Summary

Symbol	Meaning	Context
μ	Population mean	Unknown parameter we want to estimate
σ	Population standard deviation	Unknown parameter
$\bar{x}$	Sample mean	Calculated from sample data
s	Sample standard deviation	Calculated from sample data
n	Sample size	Number of observations
$Z_{\alpha/2}$	Critical Z-value	From standard normal distribution
$t_{\alpha/2,df}$	Critical t-value	From Student's t-distribution
$\hat{\theta}$	Estimate of parameter θ	e.g., $\hat{\mu} = \bar{x}$
SE	Standard error	$s/\sqrt{n}$ for means
MOE	Margin of error	Maximum likely distance from estimate to parameter
CI	Confidence interval	Range of plausible values

Decision Flowchart



"The best thing about being a statistician is that you get to play in everyone's backyard."

— John Tukey